DDN-RIKEN Collaboration: the Unified Multimodal Retriever Service

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Executive summary

Access to the **RedSQL Foreign Data Wrapper (FDW)** is essential to fully realize the capabilities of the UMRS system being developed at RIKEN. This integration is critical for enabling advanced metadata operations, improving performance, and supporting efficient, large-scale processing of multimodal scientific data on Infinia.

The system will be used in a **real-world production environment**, providing DDN with **direct technical feedback** on RedSQL's capabilities and limitations. This feedback loop could help guide future development and optimization of Infinia for AI-driven scientific workloads.

Introduction

Since August 2024, Emmanuel has been working with DDN Japan and has been assigned to RIKEN, a leading scientific research institution in Japan. His role focuses on data management in the context of AI for scientific research.

One of the key projects he is involved in is the **Unified Multimodal Retriever Service (UMRS)**. This initiative aims to provide a high-level data management service over Infinia—or any S3-compatible server—to support inference on multimodal datasets. The service is currently being deployed at RIKEN for a project that handles large-scale multimodal data generated by the SPring-8 synchrotron facility.

In this context, Emmanuel Jeannot is exploring how to leverage the low-level features of **Infinia RedSQL** to enhance metadata management, accelerate computation, streamline workflow control, and improve the retrieval of results.

RIKEN and the collaboration with DDN

RIKEN is Japan's premier public research institute, conducting cutting-edge research across a wide range of scientific disciplines, with a strong emphasis on computational science. Its Kobe site is home to **Fugaku**, which held the title of the world's most powerful supercomputer from June 2020 to June 2022 and remains among the global top ten. Fugaku is supported by DDN storage systems, and RIKEN is currently planning its successor, **Fugaku-Next**, expected to launch by the end of this decade. As part of its ongoing innovation efforts, RIKEN recently acquired an **Infinia** system for proof-of-concept testing, further strengthening its collaboration with DDN Japan—of which it is a major client.

Since August 2024, **Emmanuel Jeannot** has been working with DDN Japan and is currently seconded to RIKEN's Tokyo office, within the **AI for Science** division led by Dr. Mohamed Wahib. His research and development work focuses on improving data management strategies in AI-driven scientific workflows, particularly in the context of **multimodal data**—where integrating diverse data types enables deeper insights into complex phenomena that cannot be fully understood through a single modality.

One of the projects Emmanuel Jeannot is contributing to involves data generated by the **SPring-8 synchrotron**. This project focuses on tomography images of road core samples, which are extracted from pavement to assess road quality and inform maintenance planning. These multimodal datasets are transferred to RIKEN's Kobe site, where they are processed using the Infinia server and the Fugaku supercomputer. Various AI models—tailored to each data modality—are applied to generate embeddings, which are then stored for further post-processing. In full-scale production, the system is expected to handle and process several **petabytes of data daily**, underscoring the need for robust, scalable, and efficient data management solutions.

The Unified Multimodal Retriever Service

The **Unified Multimodal Retriever Service (UMRS)** is a high-level data management service developed by Emmanuel Jeannot as part of the SPring-8 data processing initiative. Its primary objective is to provide seamless and efficient management of multimodal data stored on S3-compatible systems—particularly the **Infinia** storage platform—by automating the generation and handling of embeddings.
UMRS offers the following key features:

- **Automated Embedding Computation**: Automatically triggers the computation of embeddings as soon as new data is ingested into the system.
- **Abstracted Metadata Management**: Utilizes a custom **PostgreSQL** database to manage embeddings and associated metadata, simplifying access and control.
- **Multimodal Data Handling**: Leverages both object metadata and storage structure to manage diverse data types effectively.
- **Advanced Embedding Storage**: Integrates **PGVector**, a PostgreSQL extension designed for efficient embedding storage, enabling high-performance operations such as similarity comparison, indexing, and querying.
- **Model-Agnostic Architecture**: Supports easy integration of various AI models, allowing the system to process data from any modality.
- **Performance-Oriented Design**: Optimized to take full advantage of available hardware resources, including GPUs, high-speed networks, and storage systems.

UMRS is designed to scale with the demands of large scientific workflows, making it a powerful tool for managing and analyzing complex, multimodal datasets in high-performance computing environments.

Toward Better Integration with Infinia through RedSQL

RedSQL is a low-level feature of the Infinia S3-compatible storage system that enables access to both system-level and user-defined metadata (e.g., tags) associated with stored objects. While RedSQL exposes a SQL-like interface, it is not a full-featured SQL engine—it lacks support for advanced operations such as joins and complex WHERE clauses.
However, by leveraging the **RedSQL Foreign Data Wrapper (FDW)** within PostgreSQL, it becomes possible to integrate RedSQL tables directly into a PostgreSQL environment. This opens the door to more powerful and flexible metadata management capabilities, which are essential for scaling and enhancing the **Unified Multimodal Retriever Service (UMRS)**. As such, the **expected benefits of RedSQL FDW Integration for UMRS are:**

- **High-Performance Metadata Access**: Enables faster and more efficient querying of object metadata, which is critical for large-scale data workflows. This includes filtering by file type, creation date, size, or custom metadata fields.
- **Tag-Based Object Management**: Allows tagging of objects and querying based on those tags, facilitating better organization, categorization, and retrieval of multimodal datasets.
- **Improved Data Integrity and Monitoring**: Enables checks to ensure that all stored objects have corresponding embeddings, helping maintain consistency and completeness in the data pipeline.
- **Smarter Data Lifecycle Management**: Facilitates rapid access to metadata needed to make decisions about data compression, deletion, or migration to secondary storage—especially important when dealing with petabyte-scale datasets.
- **Enhanced Query Capabilities**: Through PostgreSQL, users can perform complex queries, joins, and aggregations that are not natively supported by RedSQL, enabling more sophisticated data analysis and reporting.
- **Unified Metadata View**: Integrating RedSQL tables into PostgreSQL allows UMRS to present a unified view of both system and user metadata, simplifying development and reducing the need for custom logic.

- **Foundation for Automation and Policy Enforcement**: Enables the implementation of automated rules and policies (e.g., auto-tagging, access control, data expiration) based on metadata conditions.

Strategic Importance for DDN

RIKEN is a major DDN customer, and the SPring-8 data processing project offers a unique opportunity to showcase **Infinia's value in real-world, AI-driven scientific workflows**. The **UMRS** service, developed in this context, highlights how Infinia can support large-scale, multimodal data processing beyond LLMs.

Granting access to the **RedSQL FDW** would enable advanced metadata operations, improve performance, and unlock new features. Most importantly, this integration will be deployed in a **live production setting**, creating a **valuable feedback loop** where real user experience directly informs future improvements to Infinia and RedSQL.